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Blockchain-based Watermarking Approach for Securing E-healthcare Data

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Abstract. Medical images are essential for clinicians to make accurate clinical diagnoses and determine appropriate treatments. Nevertheless, the transmission and dissemination of such confidential data give rise to security and privacy concerns. To tackle this problem, the technique of image concealing is employed as an efficient method to safeguard images. This proposed work introduced an innovative method that protects digital data against copyright violations. The proposed method embeds and extracts a watermark using a combination of three techniques Redundant discrete wavelet transform (RDWT), Schur decomposition, and Multiresolution singular valued decomposition (MSVD). RDWT makes the watermarking process less sensitive to image shifts, allowing for more hidden information. Schur and MSVD are used to prove ownership and protect copyright. This method achieves a desirable balance between the visibility of the watermark and its resistance to tampering. Additionally, blockchain encryption is employed to enhance the system's security. A denoising approach is used to enhance the clarity of the retrieved watermark. Metrics, including structural similarity index measurement (SSIM), normalized correlation coefficient (NCC), and peak signal-to-noise ratio (PSNR) are used to evaluate the performance of the suggested approach. Various attacks are employed to assess the algorithm's resilience.

Keywords: Watermarking, RDWT, MSVD, Schur decomposition, Blockchain Encryption, DnCNN (Denoising convolution neural network), weiner filter denoising

1 Introduction

Rapid advancements in information technology have made it easier than ever to disseminate and exchange digital medical content like images, signals, and audio. Healthcare images hold sensitive patient data such as medical history, diagnoses, and treatments [1, 2]. Unauthorized access poses risks of privacy violations, identity theft, and patient harm. Manipulating or altering these images can lead to misdiagnoses, treatment errors, and medical mistakes [3]. These images are critical for research, collaboration, and telemedicine. Ensuring the integrity and authenticity of images is vital for accurate patient care. Safeguarding their security prevents data loss, unauthorized sharing, and compromised research findings [4–6]. For decades, researchers have worked on digital watermarking to secure digital copyrights and sensitive information efficiently [7, 8]. This technology protects copyright and prevents unlawful modification by hiding sensitive information in digital content. Blind, semi-blind, and non-blind digital watermarking are categorized according to the needs of watermark extraction. To extract or detect the watermark, the original cover image is required for non-blind watermarking. For semi-blind watermarking to work, either the original data or the watermark itself must be preserved to extract it. Digital content that has been watermarked can have its watermark extracted or detected immediately using blind watermarking.

The techniques used to insert digital watermarks also allow for their categorization into two distinct domains: spatial and transform [9]. Although spatial domain watermarking is simple to implement, it is not resistant to typical attacks. Despite this, research has focused on the transform domain due to its improved performance. In watermarking algorithms, invisibility is crucial, indicating the need for the watermarked image to resemble the original cover image to avoid detection closely. Furthermore, the strength of the watermark indicates the algorithm's capacity to endure different types of attacks without compromising its integrity. Watermarking algorithms based on the transform domain have been abundant in recent years. Overall digital image watermarking addresses various security benefits, including copyright protection, authentication of material, verification of data integrity, detection of forgeries, protection of privacy, and administration of digital rights [10, 11].

The initial challenge of digital watermarking is balancing invisibility, resilience, and capacity, and enhancing one of these qualities will compromise the others. Having these three features simultaneously is ideal for a watermarking system. Payload size measures the data hidden inside the host image, but a bigger payload reduces imperceptibility. A suitable watermarking method will strike a balance between all of these aspects.

This article presents a novel method for data concealing that combines RDWT–Schur–MSVD transforms. This method efficiently uses the advantages of multiple data concealment techniques to attain superior performance in terms of invisibility, resilience, and safeguarding. Although the data-hiding approaches discussed in the article provide authenticity and resilience against typical assaults, they suffer from reduced security due to decreased computational complexity.

This strategy seeks to surpass this constraint using blockchain-based encryption and provides enhanced outcomes. The innovations of this work can be outlined as follows:

- Advantages of the hybrid watermarking method:- A novel watermarking approach is devised that integrates the strengths of RDWT-Schur-MSVD. RDWT enhances the payload capacity due to its shift invariance property, as opposed to DWT. Schur decomposition offers advantages like decorrelation, dimensionality reduction, and resilience in the transformed domain. Wavelet-based multiresolution SVD approaches provide benefits, including localization, scalability, and resilience to compression. The outcome of this hybrid technique offers robustness, security, and safeguards against attacks.
- Security using Blockchain-based encryption:- The watermarked image is additionally safeguarded by employing a robust blockchain-based encryption technique, which gives the best results of performance parameters i.e. entropy, NPCR, UACI.
- Enhancing robustness using denoising methods:- DnCNN and a weiner filter are applied to the extracted watermark to increase visual quality. After analysis, it is noticed that DnCNN makes the watermark more robust than the Weiner filter.
- Matched features using ORB after attacks:- Experimental findings prove that features of the cover image and watermarked image after the attack give best feature matching compared to other feature extractors [12–14].
- Experimental findings and analysis:- Compared to previous existing work, the results show that the proposed work performs better in imperceptibility, robustness, and security.

This work is organized into the following sections: Section 2 discusses the literature study, section 3 outlines the proposed methodology for embedding and extracting watermarks, section 4 gives the findings and analysis of comparison, section 5 summarizes the work.

2 Literature Review

This part thoroughly reviews the current progress in digital watermarking field, which offers information on the latest methods and strategies used to protect digital image assets

Shen presented a blind embedding strategy that is based on the QZ algorithm [15]. It ensured strong imperceptibility and resisted multiple image processing attacks, such as the screenshot assault. Additionally, it is also designed to prevent forgeries and false positives. A new symmetric watermarking architecture was introduced by ref [16] that used chaotic encryption based on neural networks. The suggested architecture generated various chaotic sequences and keys for watermark encryption by simulating chaotic systems using neural networks. Tang proposed a unique two-step process for concealing medical images[17]. Using reversible data hiding, the first step is to create a QR code containing the patient's diagnosis information (PDI) and insert it into a hidden medical image. Step two involved disguising the PDI-containing hidden medical image inside a seemingly innocuous target image. The revolutionary two-stage concealment process incorporated a lossless compression method called soft compression, guaranteeing that the reconstructed secret medical image and PDI are faithful reproductions of the originals. Awasthi proposed a dual image watermarking method that takes advantage of the features of Schur decomposition and SVD, DCT and LWT for hiding to enhance the scheme's performance and make it more resilient and imperceptible[14]. The authentication process is completed by taking advantage of SURF characteristics. A reversible MIW technique is proposed by ref [4] for the secure transmission of medical pictures and EPR. The integrity of medical images, ERP secrecy, and ownership authentication are guaranteed by inserting the encoded information, tamper detection, and tamper recovery bits. With very little computing overhead, the suggested system attains excellent imperceptibility, robustness against attacks, security, tamper detection, retrieval, and payload. By merging DWT and SVD[3], author presented a hybrid resilient invisible image watermarking method. Additionally, the method offers improved resilience against assaults involving noise, filters, and image blur compared to existing works. The limitation was using adaptive algorithms to select the optimal alpha value. In ref [18], the embedding process consisted of two stages i.e. robust and reversible embedding stage. The stage of robust embedding, it commences by picking PHT moments with a high level of robustness. Subsequently, a method of adaptive normalization is presented to achieve a balance between resilience and imperceptibility. Future work is embedding a single watermark bit into many PHT moments, enhancing the watermarking system's robustness. A wavelet transform approach is suggested for image content authentication by reversible watermarking [19]. Every block goes through a wavelet transform to include the watermark information, protecting the entire image's wavelet domain against manipulation. Regarding security, two watermarks undergo encryption and scrambling by chaotic mapping using distinct keys, enhancing the watermark's security and the level of unpredictability during embedding. Ref [20] introduced an innovative method for watermarking that utilizes a blend of NSST-HD-SVD approach. The watermarked image is made more resilient and protected using this method. The technique utilizes a grayscale image as the watermark and incorporates it into numerous cover images. Applying the NSST-HD-SVD approach to the produced images yields the final watermarked images.

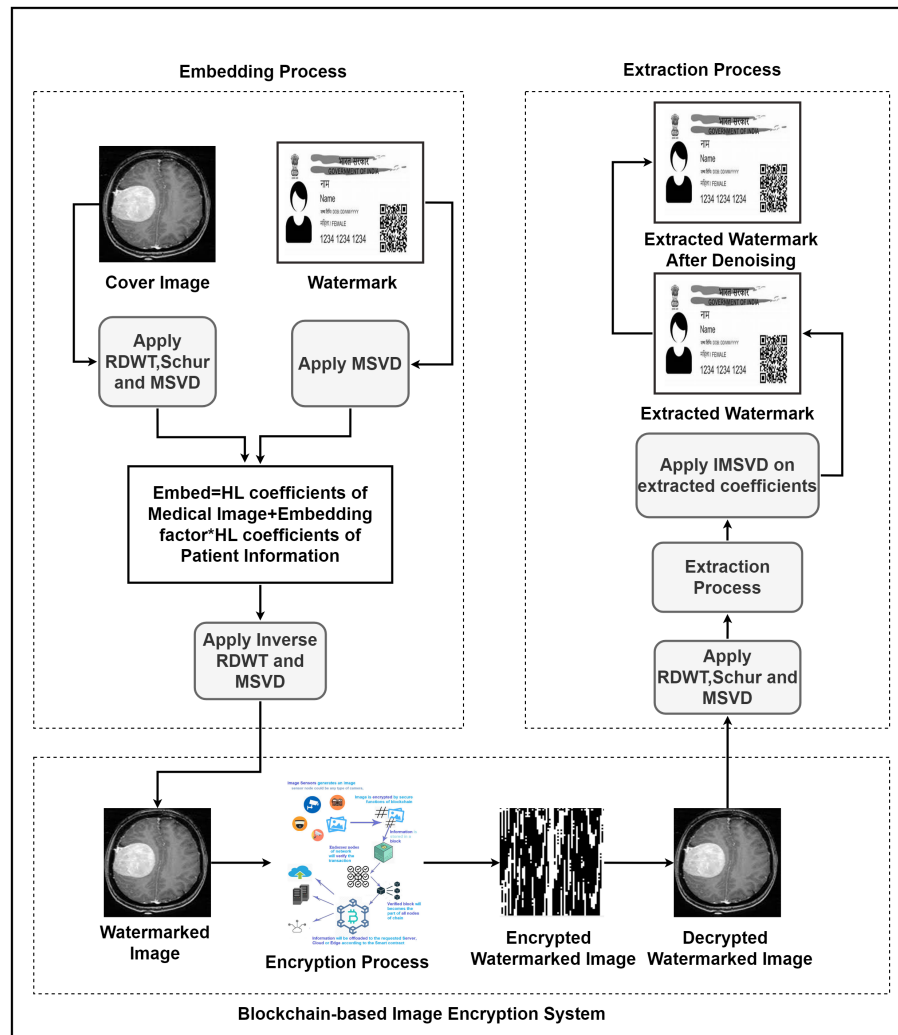


Fig. 1: Methodology diagram of proposed work

3 Proposed Methodology

This part provides an in-depth overview of the embedding and extraction process. The watermarking embedding technique involved MRI and ChestXray images of various distinct patients, each with a size of 512×512 pixels. Additionally, patient information as an initial watermark of size 512×512 is utilized. Figure 1 demonstrates the method of hiding

the watermark image inside the cover image.

In the proposed watermarking scheme, we conducted a hybrid of RDWT, Schur decomposition, and MSVD operations on medical images before the embedding process. The RDWT technique breaks down the medical image into its constituent lowfrequency and highfrequency components, namely ' Ll ', ' Lh ', ' Hl ', and ' Hh '. After getting four components, we select one component, ' Hl ', which maintains the delicate balance between robustness and imperceptibility. After this, Schur decomposition is applied on ' Hl ' which decomposed a square matrix into an upper triangular matrix ' U ' and its conjugate transpose ' T '. Finally, MSVD is applied on coefficients of ' T ' and gives ' Cy ' and ' Cu ' coefficients.

On the other side, MSVD is applied on the watermark image i.e. patient information which provides ' Wy ' and ' wU ' coefficients. During the embedding process, coefficients ' wY ' of the watermark image are integrated inside coefficients of the cover image with the best scaling factor i.e. 0.5, which balances imperceptibility and robustness. After embedding, the inverse of MSVD, Schur, and RDWT is applied to acquire the watermarked image similar to the medical cover image as given in Figure 1. The detailed steps of the embedding process are given in Algorithm 1 and This hybrid domain improves the scheme's invisibility and robustness.

The watermarked image is encrypted through blockchain-based encryption as given in figure 2, which enhances security[21]. Image sensors, a range of devices created for capturing images, start the process. The sensors send collected data to the blockchain network. Each node in this network has distinct private keys provided by a certificate authority. The images are encrypted using a specific method before being sent to endorser nodes. After approval by these nodes, the encrypted images are permanently stored in the blockchain. End-users can access particular images using the hashed transaction ID which linked to each image. The detailed steps of encryption are given in algorithm 2 and 3. After

Algorithm 1: Embedding Process of medical image Watermarking based on RDWT_Schur_MSVD Hybrid technique

```

Input : cover, ori_mark,  $\alpha$ 
Output: W_img
// Stage:1 Perform RDWT on the medical image of size '512 × 512'
1  [Ll, Lh, Hl, Hh] = RDWT(cover);
// Stage:2 Use Schur on lh Coefficients
2  [U, T] = schur(Lh);
// Stage:3 Use MSVD on Schur componnets 'T'
3  [Cy, Cu] ← MSVD(T);
// Stage:4 Use MSVD on watermark 'ori_mark' of size '512 × 512'
4  [Wy, Wu] ← MSVD(ori_mark);
// Stage:5 Embedding watermark components 'Wy' inside cover components 'Cy'
5  WatHL ← Cy.HL +  $\alpha$  * Wy.HL;
6  tmp ← Cy.HL;
7  Cy.HL ← WatHL;
// Stage:6 Apply Inverse MSVD on Watermarked Image
8  WatHL1 ← IMSVD(Cy, Cu);
// Stage:7 Perform Inverse Schur on 'WatHL1' components of Watermark Image
9  WatHL ← U * WatHL1 * U';
// Stage:8 Apply Inverse RDWT to obtain watermarked Cover Image
10 watermarked_img = RDWT(Ll, WatHL, Hl, Hh);
11 W_img ← reshape(watermarked_img);
12 return W_img;

```

Algorithm 2: Encryption process using Blockchain

```

Input: BlockchainWebService
Output: GenesisBlock(GB)
1  while Trnasaction is still valid. do
2      if node is verified == 1 then
3          if request is correlated == 1 then
4              if Node is identified as processed request == 0 then
5                  process for the response to  $C_i$ ;
6                  Hashing the image(image);
7                  Updating the chain;
8              end if
9          else
10             Response to Node that the request is not valid;
11         end if
12     end if
13 else
14     Deny the Request;
15 end if
16 Verify and append the block to the blockchain.;
17 end if
18 end while

```

getting the watermarked image, the ORB feature extractor matches the total number of features between the cover and watermarked medical even after the attack. Our primary objective is to retrieve the watermark image from the watermarked image without any degradation. The method of extraction is the opposite of the embedding method. After retrieval, DnCNN and wiener filter enhance the watermark quality, which NC measures.

4 Results and Analysis

In this section, numerical simulations are executed on the MATLAB R2022b platform, which is compatible with Windows 11, an Intel Core i5-7500 processor. The medical host images (512 × 512 pixels) i.e MRI[22] and ChestXray images[11] are obtained from kaggle for testing. For deciding the best embedding strength, table 2 displays the various performance parameter values using different embedding strengths ranging from 0.1 to 0.9. Less alpha value increases the payload capacity. A high alpha value decreases embedding capacity, which decreases the quality of the cover image. The proposed approach is simulated using a scaling factor of 0.5 to maintain the balance between imperceptibility and robustness.

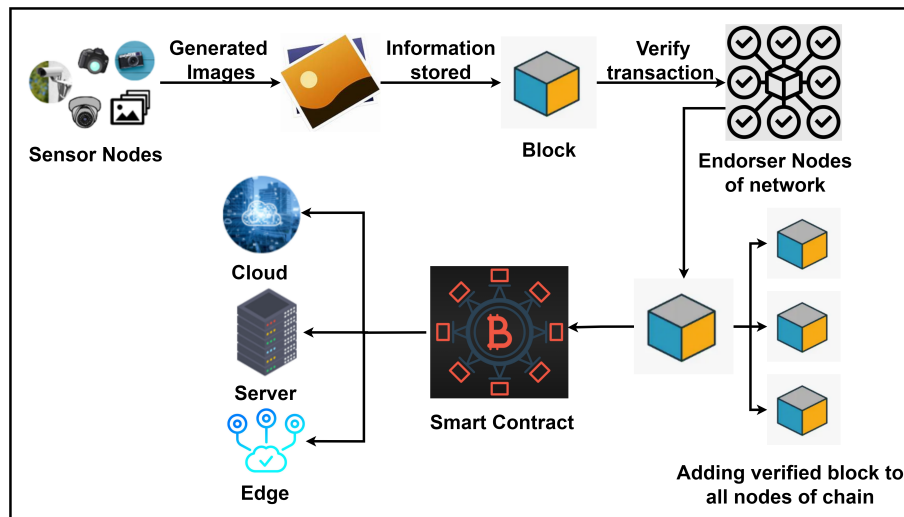


Fig. 2: Flowchart of blockchain-based encryption method

Table 3 gives the evaluation parameters results between SVD and MSVD at embedding strength 0.5. It concludes that MSVD performs better than SVD in imperceptibility, robustness, and security.

Extensive testing with several medical image types, including MRI and chest X-ray, proves that the proposed watermarking is imperceptible, safe, and robust. PSNR, SSIM, and NC are used to evaluate the efficacy of the watermarking system. The analysis of the imperceptibility in terms of PSNR and SSIM of our proposed method for different MRI and ChestXray images is given in table 4. The average PSNR and SSIM are 72.24049 and .9999957, performing better visual quality of watermarked images. Performance parameter values of security analysis also give the best result using blockchain-based encryption. Entropy, NPCR, and UACI averages are 7.999292, 0.999638, and 0.362814 in the case of MRI images, respectively. The robustness analysis regarding NC value is also given in 4. In the case of 139 MRI images, the average value of NC is .999638. In the case of 100 ChestXray images, the average value of NC is 0.99958957.

Figure 3 displays the histogram plots of two watermarked and encrypted medical images. In the given figure, Images (a) and (b) display watermarked and encrypted MRI images. Images (c) and (d) display the watermarked and encrypted chest X-ray images. It shows the color intensity fluctuation of the watermarked images before any experimentation.

Table 1: Comparison of ORB, SIFT, and SURF

Feature	ORB	SIFT	SURF
Year of Introduction	2011	1999	2006
Key Point Detector	FAST	Scale-Invariant Feature Transform (SIFT)	Speeded Up Robust Features (SURF)
Key Points	Speed and Robustness	Robustness	Speed
Descriptor	BRIEF	Vector of Locally Aggregated Descriptors (VLAD)	Histogram of Oriented Gradients (HOG)
Descriptor Length	Binary (256 bits)	128-dimensional	64-dimensional
Speed	Very Fast (2 orders of magnitude faster than SIFT)	Moderate	Fast
Orientation	Yes	Yes	Yes
Multiscale	yes	yes	yes
Patent Status	Free to Use	Patented until 2004	Free to Use
Robustness	Robust to various transformations	Robust to changes in scale, rotation, and illumination	Robust to scale, rotation, and partial occlusion
Applications	Real-time applications, robotics, mobile vision	Image stitching, object recognition, 3D reconstruction	Object tracking, image matching

Algorithm 3: Encryption Technique using Blockchain

```

Input : watermarked Image (wat_img), GenesisBlock (GB)
Output: dec_img
// Stage:1 Preprocessing
1  $[M, N] = \text{size}(\text{wat\_img});$ 
2  $Y1 = \text{uint8}(\text{zeros}(M, N));$ 
3  $r = 1;$ 
// Stage:2 Shuffling stage
4  $Sh = \text{rand}(1, 512 * 512);$ 
5  $[T, INDEX] = \text{SORT}(Sh);$ 
6  $X1 = \text{wat\_img}(:);$ 
7  $x = \text{reshape}(X1(INDEX), 512, 512);$ 
// Stage:3 Encryption Process
8 for  $p = 1$  to  $M$  do
9   for  $q = 1$  to  $32$  to  $N$  do
10     $Temp = x(p, q : q + 31);$ 
11     $y(p, q : q + 31) = \text{bitxor}(GB(r, :), X1(p, q : q + 31));$ 
12     $GB(r + 1, :) = \text{uint8}(\text{sha256hasher}.\text{ComputeHash}(Y(p, q : q + 31)));$ 
13     $r = r + 1;$ 
// Stage:3 Decryption Process
14  $GB1(1) = GB(1) + 10;$ 
15  $Y(1, 1) = Y(1, 1) + 7;$ 
16 for  $p = 1$  to  $M$  do
17   for  $q = 1$  to  $32$  to  $N$  do
18     $GB1(r + 1, :) = \text{uint8}(\text{sha256hasher}.\text{ComputeHash}(Y(p, q : q + 31)));$ 
19     $z(p, q : q + 31) = \text{bitxor}(GB1(r, :), Y(p, q : q + 31));$ 
20     $r = r + 1;$ 
// Stage:4 Inverse Shuffling
21  $z1 = z(INDEX);$ 
22  $\text{dec\_img} = \text{reshape}(z1, M, N);$ 
23 return  $\text{dec\_img};$ 

```

Table 2: Result analysis of performance parameters using the different scaling factors for MRI and ChestXray images

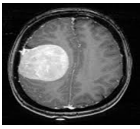
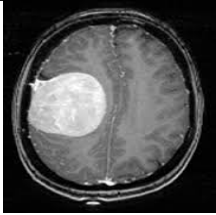
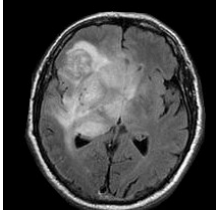
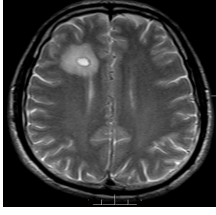
Medical Images	Alpha	PSNR	SSIM	Entropy	NPCR	UACI	NC
	0.1	Inf	1	7.9993	0.996	0.3654	0.9978
	0.3	97.545	1	7.9993	0.9961	0.3658	0.9998
	0.5	73.0889	1	7.9992	0.9959	0.3658	0.9999
	0.7	63.3443	0.9997	7.9994	0.9962	0.366	0.9999
	0.9	58.731	0.9991	7.9993	0.9961	0.3651	1
	0.1	Inf	1	7.9992	0.9963	0.3207	0.9926
	0.3	98.3162	1	7.9994	0.996	0.3201	0.9989
	0.5	73.1411	1	7.9992	0.9961	0.3203	0.9996
	0.7	63.3964	0.9997	7.9994	0.9962	0.3212	0.9998
	0.9	58.7452	0.9991	7.9992	0.996	0.321	0.9999

Table 3: Comparative analysis between SVD and MSVD

Sample Data	MRI Image		ChestXray Image	
	SVD	MSVD	SVD	MSVD
PSNR	63.9365	73.0889	61.4448	73.1411
SSIM	0.9999	1	0.9994	1
Entropy	7.9992	7.9992	7.9993	7.9992
NPCR	0.9961	0.9959	0.9959	0.9961
UACI	0.3652	0.3658	0.321	0.3203
NC	0.9359	0.9999	0.9403	0.9996

The horizontal axis of the histogram shows the intensity value. The images used grayscale with pixel values between 0 and 255. The vertical axis represents the number of pixels containing particular intensity values. It produces encrypted images by applying a blockchain-based encryption technique to watermarked images. The plots show that the information

Table 4: Average result of different performance parameters using various MRI images

Sample Images(0.5)	PSNR	SSIM	Entropy	NPCR	UACI	NC
	73.08895	0.999961	7.999221	0.999911	0.366171	0.999911
	72.49349	0.99997	7.999254	0.99983	0.318977	0.99983
	72.44849	0.999952	7.999348	0.99968	0.387787	0.99968
Avg of 139 MRI images[11]	72.24049	0.999957	7.999292	0.999638	0.362814	0.999638
	73.12019259	0.999965032	7.9992323	0.9959259	0.3086557	0.99973892
	72.66889362	0.999952151	7.999302	0.99595642	0.2998608	0.99922149
	72.48445211	0.999957523	7.9993023	0.99596024	0.3138878	0.99976548
Avg of 100 images[22]	72.7469386	0.999957892	7.9992927	0.99607643	0.3166042	0.99958957

contained within them is entirely hidden and unreadable. The cover images with a watermark are now safeguarded and secure. Images can be uploaded to the cloud for further processing without concerns about potential image misuse.

Table 5 gives the comparative analysis of both denoising techniques. It displays the NC values against various attacks. After applying the attack, the robustness of the extracted watermark is increased by DnCNN and Weiner filter. After denoising using DnCNN, all NC value is higher than 0.8 and highest NC value is .9999 against various attacks i.e. cropping, Rotation, scaling attack and Gaussian Low pass filter. On the other side, denoising using Weiner filter gives all NC value higher than 0.9 and the best NC value is 0.9986. After analysis, It has been noted that the proposed technique is resistant to the attacks that have been examined. Table 6 displays the total number of matched features between original and watermarked MRI and ChestXray images using the ORB feature extractor.

The proposed method is compared with the most advanced techniques, and encouraging results are observed. Finally, gives the best results in most of the cases compared to other existing works[14] and [23] as given in table 7. The embedding and extraction times are 0.2245s and 0.3424s, respectively. PSNR value is also higher i.e. 73.0889, than

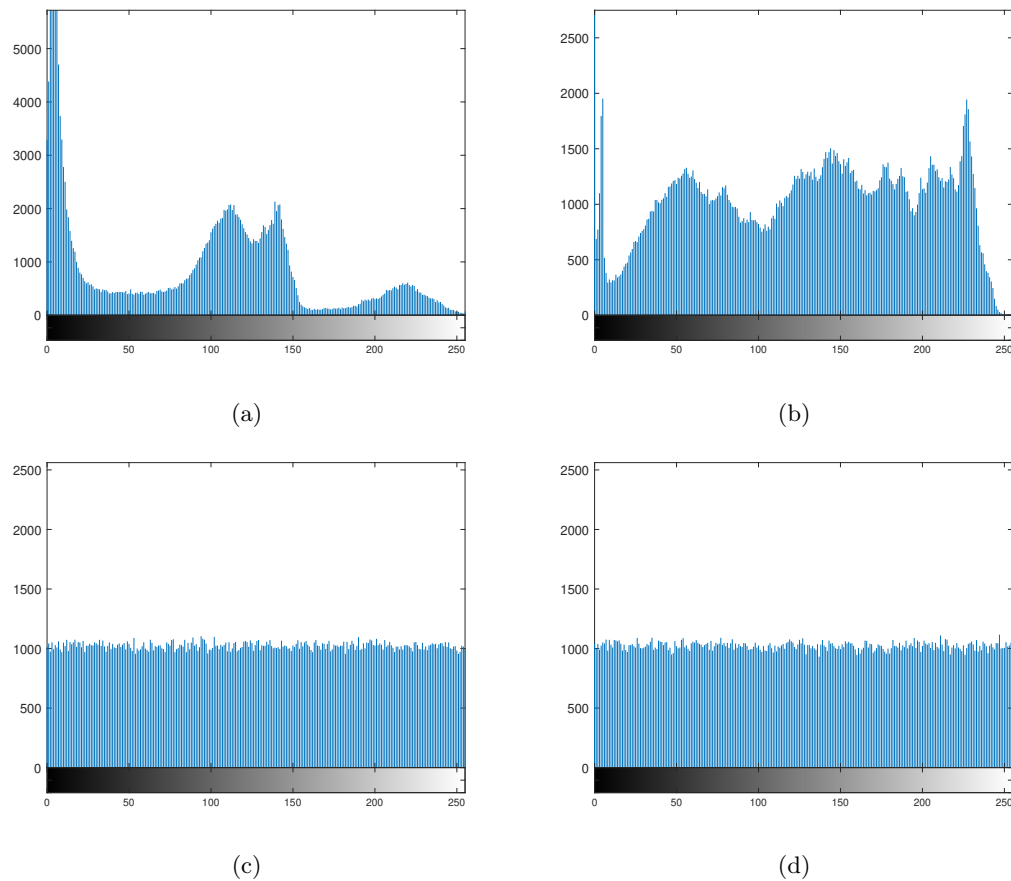


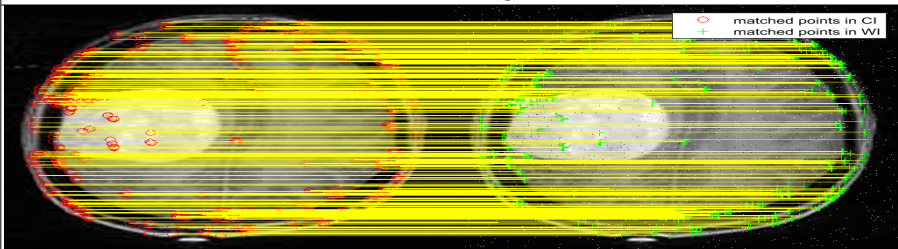
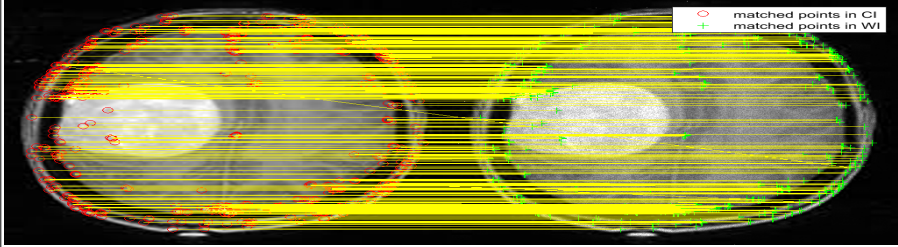
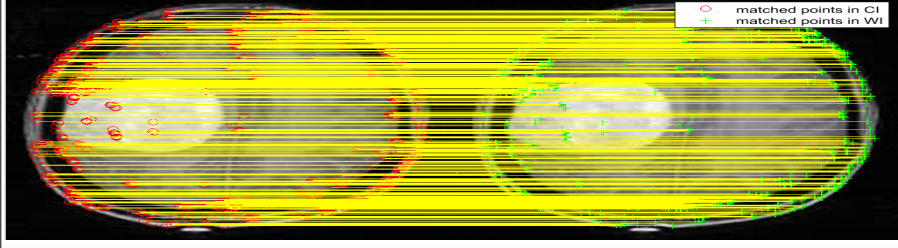
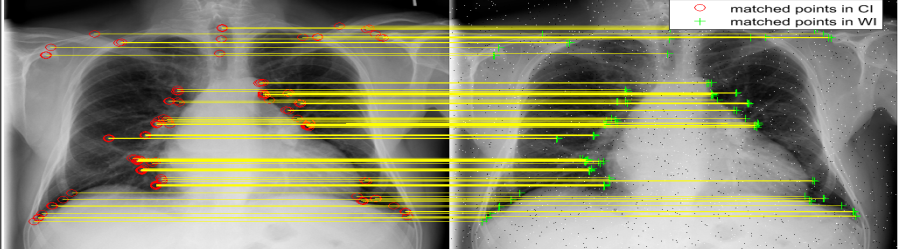
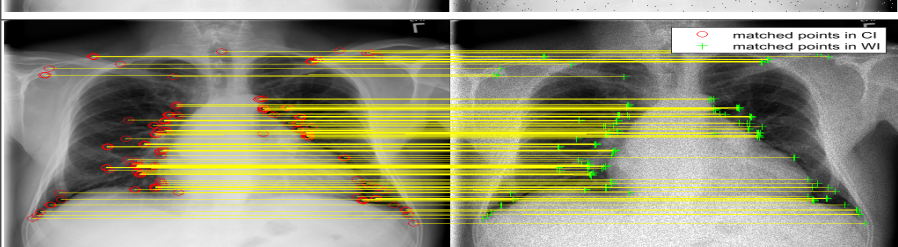
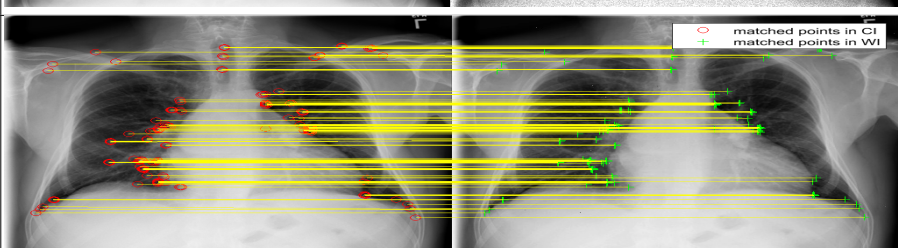
Fig. 3: a,b represents Histograms of MRI and ChestXray watermarked Images, c,d represents histograms of encrypted MRI and ChestXray watermarked images

Table 5: Improved robustness results against different attacks using different denoising techniques

Attacks	Intensity	NC(DnCNN)	NC(Weiner)
Salt and Pepper	0.01	0.9965	0.9981
	0.1	0.97	0.9884
	0.5	0.8869	0.9375
Gaussian Noise	0.01	0.9931	0.9975
	0.1	0.9512	0.9779
	0.5	0.8916	0.9406
Crop	[20 20 400 480]	0.9999	0.9986
Rotation	0.1	0.9999	0.9986
Gaussian Low Pass Filter	0.5	0.9999	0.9986
Sharpening	0.1	0.9996	0.9985
	0.5	0.9996	0.9985
Median filter	[7 7]	0.9999	0.9986
	[3 3]	0.9999	0.9986
Histogram		0.9994	0.9985
Speckle Noise	0.01	0.9985	0.9984
	0.1	0.9877	0.9961
	0.5	0.9534	0.9784
Scaling Attack	0.75	0.9999	0.9986
	0.9	0.9999	0.9986

comparative approaches. After experimental and comparative analysis, the objective evaluation proves that the suggested approach can effectively use cloud computing to deploy medical image watermarking for e-healthcare applications.

Table 6: ORB feature matching against different attacks using MRI and ChestXray image

Attacks	Matched Images	Matched Features
		922
Salt and Pepper Noise (0.01)		660
Speckle Noise (0.01)		1123
JPEG compression (Q=90)		136
Salt and Pepper Noise (0.01)		169
Speckle Noise (0.01)		207
JPEG compression (Q=90)		

5 Conclusion

In smart healthcare applications that use real-time healthcare services, a reversible medical image watermarking technique is proposed to transmit medical images and EPR securely. This article presented a Blockchain-based hybrid watermarking technique for copyright protection, security, and authentication. The proposed approach uses an embedding and extraction process, comprehensively analyzing the algorithm's performance. The embedding uses a hybrid combination of RDWT-Schur-MSVD, which gives a watermarked image. RDWT offers shift invariance property, which

Table 7: Comparison of proposed approach with existing work using different performance parameters

Attacks	Ref [14]	Ref [23]	Proposed
Histogram	0.9861	1	0.9994
Sharpening	0.9876	1	0.9996
Speckle noise (0.02)	0.9984	1	0.9985
Gaussian noise(.001)	0.9994	1	0.9998
Median Filter(3 3) ,[7 7]	0.9999	1	0.9999
Rotation	0.9958	1	0.9999
Salt and Pepper noise(0.001)	0.9999	1	0.9995
Gaussian LPF	0.9999	1	0.9999
JPEG compression(90)	0.9999	1	0.9999
Scaling(.1)	/	1	0.9999
Crop	/	1	0.9999
PSNR	45.8168	70.26	73.0889
SSIM	0.9969	0.9999	1
NC	1	0.9491	0.9999
Watermark embedding	/	0.3958	0.224493
Watermark extraction	/	0.4721	0.342396
Encryption	/	0.4366	1.131781
decryption	/	0.3181	1.035916
UACI	/	0.3587	0.3658
NPCR	/	0.9774	0.9959

helps increase the hiding capacity. Schur and MSVD are used to assert ownership and provide copyright protection. It also provides the desired balance between the robustness and imperceptibility of the watermark. Blockchain encryption also enhances the security of the system. In the extraction process. The same combination i.e. RDWT, Schur, and MSVD is applied to the watermarked image to obtain the hidden mark. The denoising technique also increases the visual quality of the extracted watermark. The experimental analysis shows that our proposed technique retains a reasonable image quality and is resistant to effective attacks as compared to other existing works. Future research will focus on evaluating the performance of color medical images, enhancing robustness and security, and identifying possible threats for in-depth investigation.

Declarations

Conflict of interest: The authors declare that they have no conflict of interest.
Ethical approval: This article does not contain any studies with human participants or animals performed by any of the authors.

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